

The Clumpy Dust Medium in MoCafe

Abstract

MoCafe's clumpy-medium mode (`par%grid_type = 'clump'`) replaces the Cartesian density grid with a population of N_{cl} non-overlapping spherical dust clumps of radius r_c placed inside a sphere of radius R ; the inter-clump space is vacuum. The clumps are discrete spheres (not a two-phase or continuous medium), implemented with the placement and ray-tracing machinery ported from LaRT but stripped to grey dust radiative transfer: because the dust opacity is frequency-independent, no Doppler width, Voigt profile, frequency tracking, or velocity field is needed, and the clump subsystem reduces to pure geometry plus a per-clump scalar opacity. The scattering and peeling-off physics are shared verbatim with the Cartesian dust path; only the optical-depth integration differs. This memo records the model, the clump-count and opacity calibration, the Random-Sequential-Addition placement, the CSR/DDA accelerated ray-tracer, and the validation against a homogeneous sphere.

1. Model

A point or extended source illuminates N_{cl} spherical dust clumps, each of radius r_c (possibly varying with clump-center radius via a profile) and grey dust opacity κ_c (extinction per unit path length). Clump centers lie in the spherical shell $r_{\text{min}} \leq r \leq R$, where $R = \text{par}\%r_{\text{max}}$ and $r_{\text{min}} = \max(0, \text{par}\%r_{\text{min}})$; with $r_{\text{min}} = 0$ the shell is a full ball. Outside the clumps the medium is vacuum, so a photon alternates between *vacuum flight* (find the next clump it enters) and *clump traversal* (accumulate $\kappa_c \ell$ along the chord of length ℓ). No opacity, and hence no interaction, occurs between clumps.

2. Clump count

For uniform clumps the realized covering and volume filling factors of N_{cl} clumps in the shell are

$$f_{\text{vol}} = \frac{N_{\text{cl}} r_c^3}{R^3 - r_{\text{min}}^3}, \quad f_{\text{cov}} = \frac{3}{4} \frac{N_{\text{cl}} r_c^2}{R^2 + R r_{\text{min}} + r_{\text{min}}^2}, \quad (1)$$

so that $f_{\text{vol}} = \frac{4}{3} f_{\text{cov}} r_c / R$ for $r_{\text{min}} = 0$. The user specifies exactly one of `clump_N_clumps`, `clump_f_vol`, or `clump_f_cov`; the other two follow from (1). When a radial *number* profile $n_{\text{cl}}(r) \propto \text{shape}(r)$ is active, N_{cl} , f_{vol} , and f_{cov} are obtained by quadrature of $n_{\text{cl}}(r)$ over the shell (the line-of-sight definition of f_{cov}). Reliable random placement requires $f_{\text{vol}} \lesssim 0.35$ (three-dimensional jamming ≈ 0.38); MoCafe aborts with guidance if the placement stalls.

3. Per-clump opacity

The grey dust opacity κ_c of a clump is set, in priority order, by one of

$$\kappa_c = \frac{\tau_0}{r_c} \quad (\text{clump_tau0}), \quad \kappa_c = n C_{\text{ext}} \ell_{\text{cm}} \quad (\text{clump_ndust or clump_nH}), \quad (2)$$

where τ_0 is the dimensionless dust optical depth from a clump's center to its surface, $C_{\text{ext}} = \text{par}\%c_{\text{ext_dust}}$ is the dust extinction per H, and $\ell_{\text{cm}} = \text{par}\%distance2cm$ converts code length to cm (the same $n C_{\text{ext}} \ell_{\text{cm}}$ used by the Cartesian grid). Alternatively the population is calibrated to a system-level radial target τ_{max} or τ_{homo} by back-solving

$$\kappa_c = \frac{\tau_{\text{target}}}{\text{GF}}, \quad \text{GF} = \frac{N_{\text{cl}} r_c^3}{R^2 + R r_{\text{min}} + r_{\text{min}}^2} \quad (\text{uniform}), \quad (3)$$

with the profile-mode geometric factor obtained by quadrature.

System scalars. From the placed population MoCafe reports

$$\tau_{\text{homo}} = \sum_i \frac{\kappa_c^{(i)} r_c^{(i)3}}{R^2 + R r_{\text{min}} + r_{\text{min}}^2}, \quad (4)$$

$$\tau_{\text{max}} = \sum_i \frac{\kappa_c^{(i)} r_c^{(i)3}}{3 \max(d_i^2, r_c^{(i)2})}, \quad (5)$$

where $d_i = |\mathbf{r}_i|$ is the clump-center distance from the origin. Equation (4) is the opacity of the clumps smeared uniformly over the shell and traversed radially; (5) is the small-angle expected radial optical depth (the max caps the singularity for a clump straddling the origin). In the uniform-isotropic limit both reduce to $N_{\text{cl}} r_c^3 \kappa_c / (R^2 + R r_{\text{min}} + r_{\text{min}}^2) = \frac{4}{3} f_{\text{cov}} \tau_0$ (the uniform discrete-clump closed form).

4. Placement: Random Sequential Addition

Clump centers are drawn one at a time and accepted only if the new clump does not overlap any previously placed one (RSA). A linked-list hash grid with cells of side $\geq 2r_c^{\text{max}}$ makes the overlap test $O(1)$: a trial center is checked only against clumps registered in its own cell and the 26 neighbors (the 27-cell rule). Uniform placement uses box rejection inside the shell; profile placement draws the center radius from the tabulated inverse CDF of $n_{\text{cl}}(r)r^2$ and isotropic (or biconical) angles. Placement runs once on the global root rank and is broadcast to one rank per node, then shared within the node through MPI-3 shared memory, so every node holds an identical clump layout.

5. Accelerated ray-tracing

CSR acceleration grid. A uniform C^3 grid over the bounding box indexes the clumps in compressed-sparse-row form: a two-pass build fills `cg_start` (per-cell offsets) and `cg_list` (clump indices), registering each clump in every cell its bounding box overlaps. The grid and the clump arrays live in MPI-3 shared memory (one copy per node).

Ray–sphere intersection. For a ray $P(t) = \mathbf{o} + t \hat{\mathbf{k}}$ and a clump centered at $\mathbf{c}$ with radius r_c , writing $\mathbf{b} = \mathbf{o} - \mathbf{c}$,

$$t_{\pm} = -(\mathbf{b} \cdot \hat{\mathbf{k}}) \pm \sqrt{(\mathbf{b} \cdot \hat{\mathbf{k}})^2 - |\mathbf{b}|^2 + r_c^2}, \quad (6)$$

with a real intersection when the discriminant is non-negative and $t_+ > 0$.

DDA traversal. The next clump a photon in vacuum will enter is found by stepping the ray through the CSR grid with the Amanatides & Woo (1987) digital differential analyzer: the per-axis distances t_x, t_y, t_z to the next cell face are advanced by $\Delta t_x = \Delta x / |k_x|$ (and similarly for y, z); at each cell every registered clump is tested with the ray–sphere formula and the nearest positive entry is kept. Traversal stops once the cell-entry distance exceeds the best hit or the sphere boundary.

Transport. `raytrace_to_tau_clump` advances the photon until a sampled optical depth τ is accumulated: inside a clump it scatters at $ds = \tau_{\text{rem}}/\kappa_c$ if $\tau_{\text{rem}} \leq \kappa_c \ell$, otherwise it subtracts $\kappa_c \ell$, steps to the clump exit, and resumes in vacuum; in vacuum it advances to the next clump (or escapes through the sphere). `raytrace_to_edge_clump` sums $\kappa_c \ell$ over every clump the line of sight crosses to the sphere boundary, and is used by the peeling-off estimator.

6. Reuse of the dust physics

Clump mode swaps only the two ray-tracing procedure pointers (`raytrace_to_tau`, `raytrace_to_edge`); the forced first scattering, the Henyey–Greenstein (or Mueller-matrix) scattering, and the peeling-off to each observer are *identical* to the Cartesian dust path. This is possible because the peel routines integrate the optical depth to the observer through the `raytrace_to_edge` pointer rather than calling the Cartesian routine directly; the default binding keeps the Cartesian results bit-identical. The forced-first-scattering weight $1 - e^{-\tau_0}$ uses the clump optical depth τ_0 to the sphere edge along the emission direction, which is zero (the photon escapes unscattered) for sight lines that miss every clump. The birth clump is identified at emission so a source inside a clump is handled correctly.

7. File I/O and the standalone generator

With `par%save_clump_info` the population is written to a binary table (FITS or HDF5): columns X , Y , Z (centers) plus `R_CLUMP` and `RHOKAP` when non-uniform, and header keywords `N_CLUMPS`, `SPHERE_R`, `RMIN`, `CL_RAD`, `F_VOL`, `F_COV`, `TAU0`, `RHOKAP`, `RMAX`, `DISTUNIT`, `DIST_CM`. A pre-built population is read back through `par%clump_input_file` (a `DENSITY/DENS` column is converted to opacity with $C_{\text{ext}} \ell_{\text{cm}}$), optionally rescaled so its realized τ_{max} or τ_{homo} matches a target. The Python helper `python/make_clumps.py` writes such files (uniform random populations or a single centered clump).

8. Validation

Homogeneous limit. A single clump centered at the origin with $r_c = R$ and $\kappa_c = \tau_0/R$ fills the sphere with uniform opacity, so it must reproduce the homogeneous dust sphere of radial optical depth τ_0 . At $\tau_0 = 1$, albedo 0.5, isotropic scattering, 2×10^6 photons, the integrated images agree with the Cartesian reference to Monte-Carlo precision:

section	homogeneous	giant clump	rel. diff.
Scattered	6.38690×10^1	6.38682×10^1	−0.001%
Direct	1.21791×10^2	1.21791×10^2	+0.000%
Direct0	3.31062×10^2	3.31062×10^2	+0.000%

The direct images are deterministic (one source point) and match exactly; the scattered totals match to 10^{-5} . The mean number of scatterings agrees to 0.06%. More generally the clumpy result approaches the smooth sphere as f_{cov} grows at fixed τ_{homo} .

9. Scope

Gas velocity and temperature are out of scope at this stage, so the LaRT Doppler width, Voigt profile, frequency tracking, per-clump velocity, and the Lyman- α resonance machinery are not ported. The ray-tracer assumes non-overlapping clumps (RSA enforces this); the overlap-aware path is recorded for a future revision. See `AMR_CLUMPS_PLAN.md` Part B.

References

- [1] Amanatides, J., & Woo, A. 1987, A Fast Voxel Traversal Algorithm for Ray Tracing, in Eurographics '87, 3